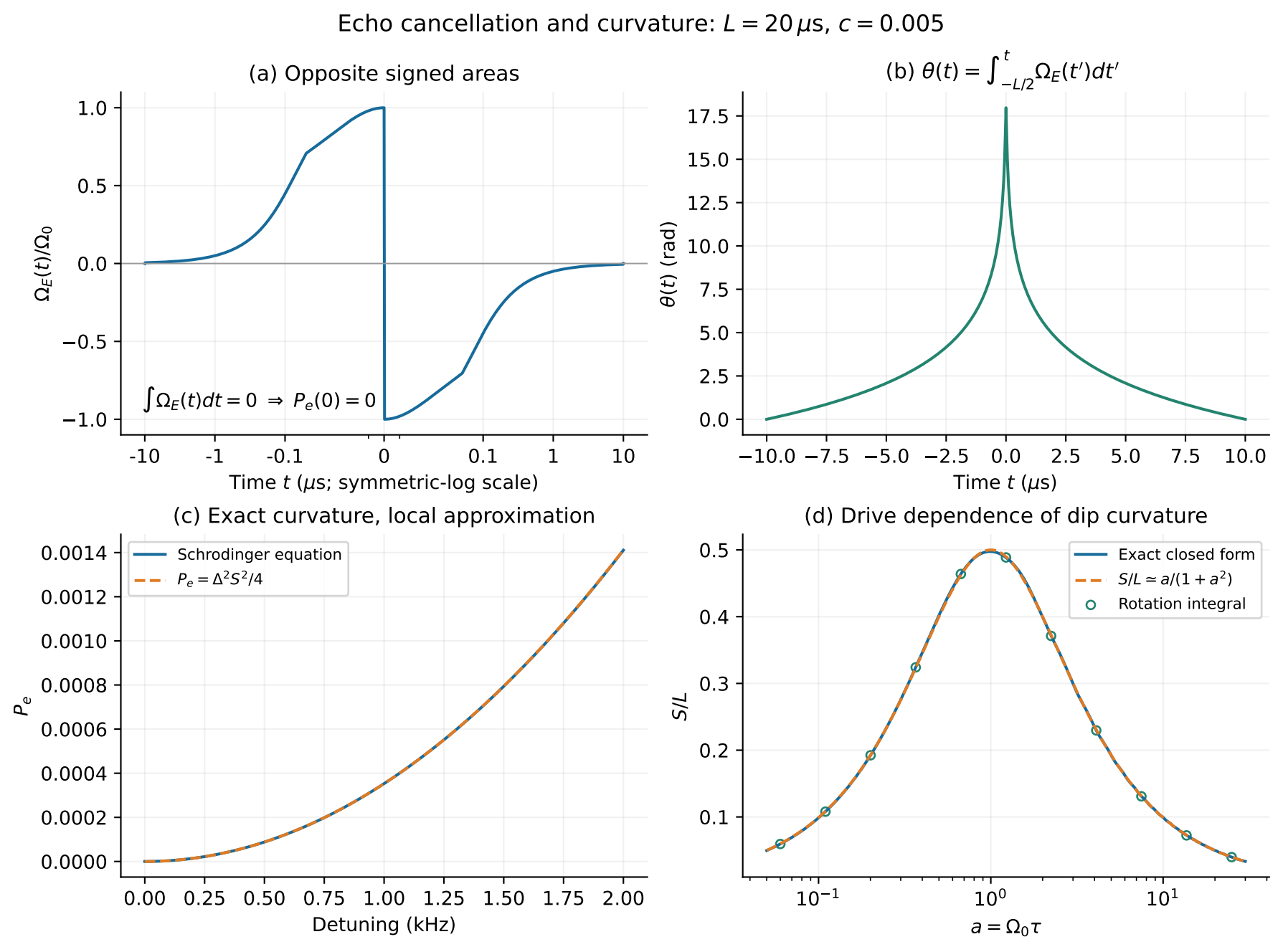
Echo cancellation and curvature: $L = 20\,\mu$s, $c = 0.005$
(a) Opposite signed areas
$\Omega_E(t)/\Omega_0$
$\int \Omega_E(t)dt = 0 \Rightarrow P_e(0) = 0$
Time t (μs; symmetric-log scale)
(b) $\theta(t) = \int_{-L/2}^{t} \Omega_E(t')dt'$
$\theta(t)$ (rad)
Time t (μs)
(c) Exact curvature, local approximation
Schrodinger equation
$P_e = \Delta^2 S^2/4$
P_e
Detuning (kHz)
(d) Drive dependence of dip curvature
Exact closed form
$S/L \simeq a/(1+a^2)$
Rotation integral
S/L
$a = \Omega_0 \tau$